BUS123 · MATH M03 · L01

Pre-Reading

Discounts: Trade and Cash

Gerrish School of Business

Connecting to What You Know

In our last module, you worked with percentages — converting between decimals and percents, calculating percentage change, and finding amounts from percentages. Today's material builds directly on those skills. Discounts are simply percentages applied to prices, and once you can work with percentages confidently, discount math follows naturally.

You have probably encountered discounts as a consumer: "30% off" a jacket, or "buy within 10 days and save 2%." This pre-reading gives you the vocabulary and formulas to work with discounts the way businesses do — precisely and systematically.

Part 1: Trade Discounts

When a manufacturer or wholesaler publishes a **list price** (also called a catalog price), they rarely expect every buyer to pay that full amount. Businesses at different points in the supply chain receive different reductions from the list price, called **trade discounts**. A retailer might receive 30% off list. A large chain might receive 40%. The actual amount the buyer pays is called the **net price**.

The core relationship:

To find the net price, we use the **complement** of the discount rate. The complement is simply (1 minus the discount rate). If the trade discount is 30%, the complement is 0.70 — meaning the buyer pays 70% of the list price.

```
Net Price = List Price × (1 – Discount Rate)
```

Worked Example — Tidal Goods Co.:

Tidal Goods Co. orders wetsuits from a supplier. The list price is $189.99 per unit, and Tidal Goods receives a 30% trade discount as a retail partner.

- Complement: 1 – 0.30 = 0.70
- Net Price: $189.99 × 0.70 = **$132.99**
- Discount Amount: $189.99 – $132.99 = $57.00

Why the complement works:

If you receive 30% off, you pay the remaining 70%. Multiplying by 0.70 gives you the net price in one step, rather than calculating the discount amount first and then subtracting it. Both approaches give the same answer — the complement method is faster.

Part 2: Chain Discounts

Sometimes a supplier offers more than one trade discount on the same item. For example, a standard trade discount might be 20%, with an additional 10% for large orders and another 5% for preferred accounts. When multiple discounts apply, they are called **chain discounts** and written as "20/10/5."

The critical rule: **each discount applies to the result of the previous discount**, not to the original list price. This means you cannot simply add the rates together. 20/10/5 is not the same as a single 35% discount.

Chain Discount Formula:

```
Net Price = List Price × (1–d1) × (1–d2) × (1–d3)
```

Worked Example — Surfboard Fins:

Tidal Goods orders surfboard fins with a list price of $64.00 per set. The supplier offers a 20/10/5 chain discount.

- Step 1: $64.00 × (1 – 0.20) = $64.00 × 0.80 = $51.20
- Step 2: $51.20 × (1 – 0.10) = $51.20 × 0.90 = $46.08
- Step 3: $46.08 × (1 – 0.05) = $46.08 × 0.95 = **$43.78**

Common Mistake — Adding Rates:

Adding 20 + 10 + 5 = 35% would give $64 × 0.65 = $41.60 — a different (lower) answer. The chain method gives $43.78. These are not equal because each discount is applied to an already-reduced base, not the original list price.

Part 3: Cash Discounts and Payment Terms

While trade discounts reduce the price of goods, **cash discounts** reward buyers who pay their invoices early. Suppliers offer cash discounts to improve their own cash flow — getting paid sooner is worth a small reduction in the amount received.

Cash discount terms are written in a standardized shorthand. The most common format is:

```
2 / 10 net 30
```

2%	10 days	30 days
Cash discount rate	Discount period — pay within this window to earn the discount	Full payment due — the total credit period

Worked Example — Tidal Goods Invoice:

Tidal Goods receives an invoice for $4,200.00 from their wetsuit supplier. Terms: 2/10 net 30.

- Discount Amount: $4,200 × 0.02 = $84.00
- Payment if paid within 10 days: $4,200 × (1 – 0.02) = $4,200 × 0.98 = **$4,116.00**
- If Tidal Goods waits past 10 days: full $4,200 is due by day 30.

The True Cost of Skipping a Cash Discount:

Not taking a 2/10 net 30 discount is equivalent to borrowing money at roughly 37% annual interest. The $84 "savings" from waiting buys only 20 extra days of payment deferral. For most businesses, it nearly always makes financial sense to take the cash discount when cash is available.

Formula Reference

Concept	Formula	Example
Net Price (single discount)	`= List × (1 – d)`	$189.99 × 0.70 = $132.99
Discount Amount	`= List × d`	$189.99 × 0.30 = $57.00
Complement	`= 1 – d`	1 – 0.30 = 0.70
Net Price (chain)	`= List × (1–d1) × (1–d2) × (1–d3)`	$64 × 0.80 × 0.90 × 0.95
Single Equiv Rate (chain)	`= 1 – [(1–d1)(1–d2)(1–d3)]`	1 – 0.684 = 31.6%

Concept	Formula	Example
Cash Disc Payment	`= Invoice × (1 − cash rate)`	$4,200 × 0.98 = $4,116
Cash Disc Savings	`= Invoice × cash rate`	$4,200 × 0.02 = $84

Check Your Understanding

Answer these questions before class. Bring your work — we'll use these as discussion starters.

1.

A surfboard has a list price of $525.00. A retailer receives a 22% trade discount. What is the net price?

2.

Tidal Goods receives a chain discount of 25/10 on a $380 order of accessories. What is the net price?

3.

An item has a list price of $75.00 and is subject to a 30/15/5 chain discount. Calculate the net price.

4.

What is the single equivalent discount rate for a 30/15/5 chain discount?

5.

An invoice for $6,500 has terms of 2/10 net 30. How much is saved by paying within 10 days?

6.

Tidal Goods receives invoice terms of "3/15 net 45." What does each number represent?

7.

A buyer can take a 2/10 net 30 cash discount or pay the full amount in 30 days. If the invoice is $2,800, what is the annualized cost of NOT taking the discount?

Answer Key

1. $525.00 × (1 – 0.22) = $525.00 × 0.78 = $409.50

2. $380 × (1 – 0.25) × (1 – 0.10) = $380 × 0.75 × 0.90 = $256.50

3. $75 × 0.70 × 0.85 × 0.95 = $42.53

4. $1 - (0.70 \times 0.85 \times 0.95) = 1 - 0.5673 = 43.3\%$

5. $\$6,500 \times 0.02 = \$130.00$

6. 3% discount if paid within 15 days; full amount due in 45 days

7. Cost = $(\$56 \div \$2,744) \times (365 \div 20) \approx 37.2\%$ annualized

Key Vocabulary

List Price

The published catalog or manufacturer's suggested price before any discounts are applied.

Trade Discount

A reduction from list price offered to buyers in the supply chain (retailers, wholesalers, etc.).

Net Price

The actual price paid after all trade discounts have been subtracted from list price.

Complement

The portion of the price the buyer actually pays; calculated as (1 – discount rate).

Chain Discount

A series of successive trade discounts applied one after the other to the same item.

Single Equivalent Rate

A single discount rate that produces the same net price as a chain of discounts.

Cash Discount

A percentage reduction offered to buyers who pay their invoice within a specified early payment window.

Payment Terms

The notation describing a cash discount — e.g., 2/10 net 30 means 2% off if paid within 10 days, full amount due in 30 days.

Discount Period

The number of days within which a buyer must pay to qualify for the cash discount.

Net Period

The total number of days a buyer has to pay the full invoice amount.